Magnetic field generation

Magnetic fields around us

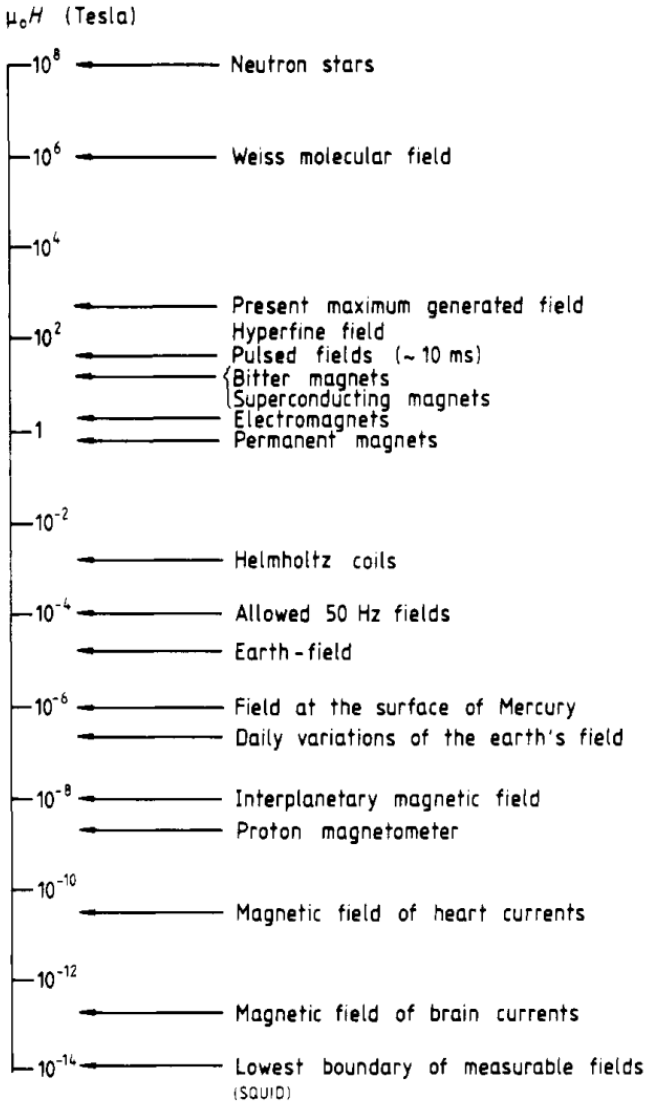


Figure 1. Scale of magnetic fields.

J. Phys. D: Appl. Phys., **15** (1982) 1545–1608.

Choice of magnets

Either you need to answer the following questions:

What field is needed?

How homogeneous the field should be? What is the sample size? What is the sample holder size?

Do you need to change the value of the field?

Do you need to change the direction of the field? How many axes?

What temperature is needed?

How fast you can sweep the field (instruments/heating/physics)?

What access do you need to the sample (wires, optics, X-rays, neutrons, …)

Can you sacrifice the sample?

Or you live with whatever you have

Units

Magnetomotive Force—The quantity of magnetic field force, or "push." Analogous to electric voltage (electromotive force).
Field Flux—The quantity of total field effect, or "substance" of the field. Analogous to electric current.
Field Intensity—The amount of field force (mmf) distributed over the length of the electromagnet. Sometimes referred to as *Magnetizing Force*.
Flux Density—The amount of magnetic field flux concentrated in a given area.
Reluctance—The opposition to magnetic field flux through a given volume of space or material. Analogous to electrical resistance.
Permeability—The specific measure of a material's acceptance of magnetic flux, analogous to the specific resistance of a conductive material (ρ), except inverse (greater permeability means easier passage of magnetic flux, whereas greater specific resistance means more difficult passage of electric current).

Quantity	Symbol	Unit of Measurement and abbreviation		
		CGS	SI	English
Field Force	mmf	Gilbert (Gb)	Amp-turn	Amp-turn
Field Flux	Φ	Maxwell (Mx)	Weber (Wb)	Line
Field Intensity	H	Oersted (Oe)	Amp-turns per meter	Amp-turns per inch
Flux Density	B	Gauss (G)	Tesla (T)	Lines per square inch
Reluctance	ℜ	Gilberts per Maxwell	Amp-turns per Weber	Amp-turns per line
Permeability	μ	Gauss per Oersted	Tesla-meters per Amp-turn	Lines per inch-Amp-turn

Quantity	CGS unit	SI Unit
Magnetic field	H = 1 Oe	$H = 1000/4\pi$ A.m^{-1}
Magnetic induction	B = 1 gauss	$B = 1 \times 10^{-4}$ Tesla
Magnetization	M = 1 erg.Oe^{-1}.cm^{-3}	$M = 1 \times 10^{3}$ A.m^{-1}

Infinite source of confusion for some (many) of us

Permanent magnets

Alnico

Ferrites

Neodymium-iron-boride

Samarium-cobalt

Rarely used as field-generation device in physics labs.

Possibly can be used in field instruments for geophysicists, agronomists, …

< 1 Tesla

temperature and history dependent

difficult to change field

cheap

compact

can be easily shaped

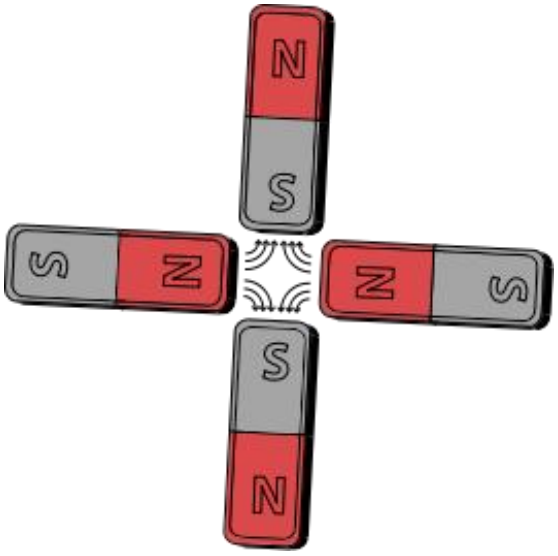
Alnico: **Al** 8-12%, **Ni** 14-26% **Co** 5-38 %, **Cu** 3-6 %, **Ti** 0-8 %, balance **Fe** (different grades, identified in 1931 [Japan], Curie temperature 700-860° C)

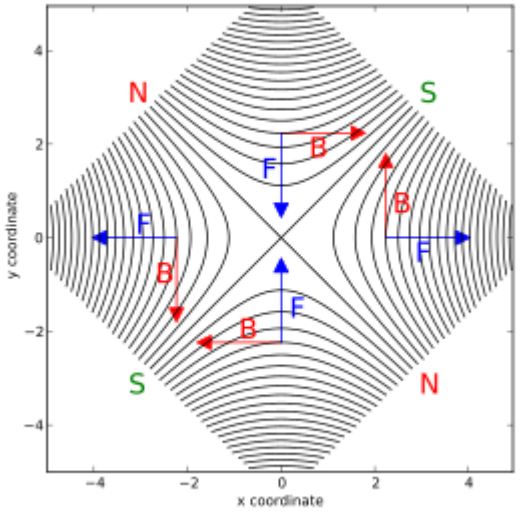
Ferrites: Fe_2O_3 blended with stuff, huge family

Neodymium-iron-boride: $Nd_2Fe_{14}B$ and derivatives (Curie temperature ~320° C)

Samarium-cobalt: $SmCo_5$ and derivatives (Curie temperature 700-800° C)

Permanent magnets – complex field profiles





quadrupole magnet (focusing electron beam)

Permanent magnets

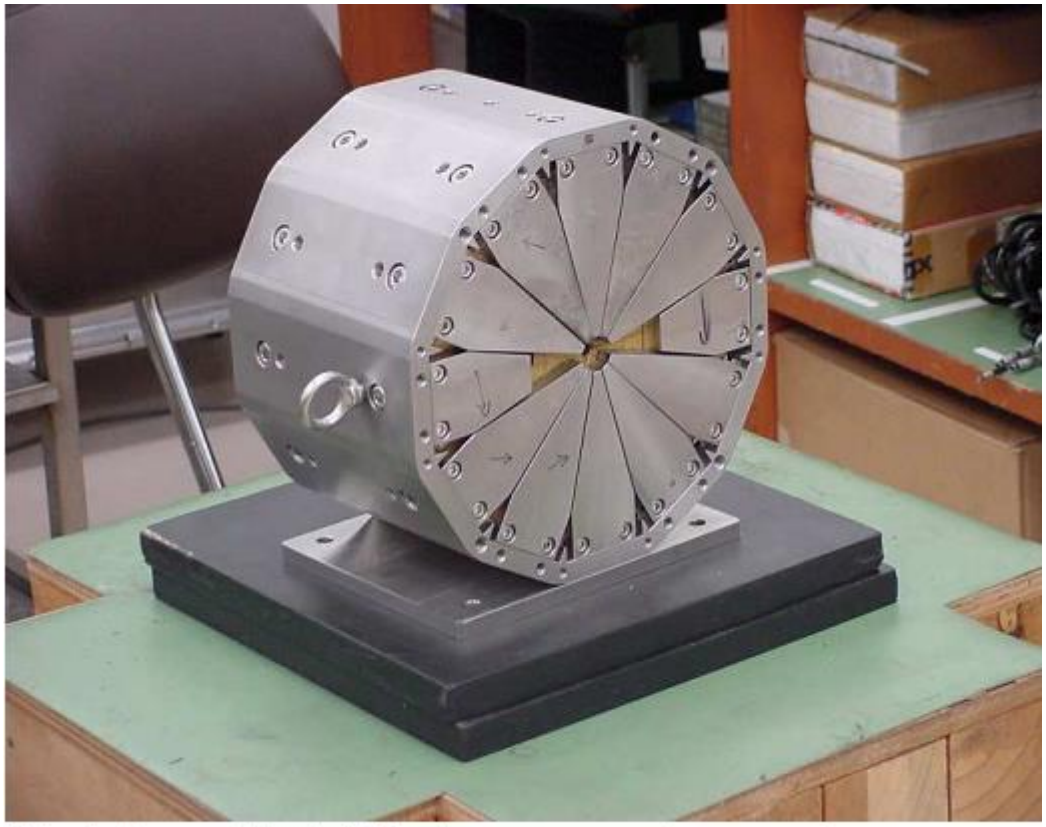
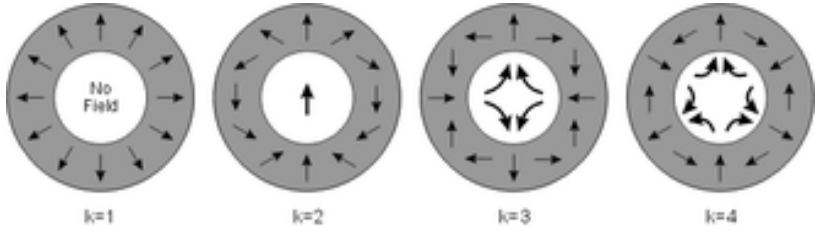


Fig.1 Assembled 4 Tesla permanent magnet



Halbach cylinder

Proceedings of the 2001 Particle Accelerator Conference, Chicago

DEVELOPMENT OF 4 TESLA PERMANENT MAGNET

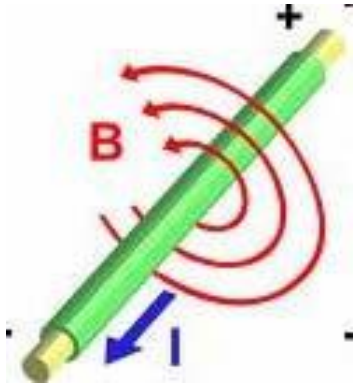
M. Kumada, T.Fujisawa, Y.Hirao, National Institute of Radiological Sciences,
4-9-1 Anagawa, Inage, Chiba, 263-8555 Japan
M. Endo, M.Aoki, T.Kohda, Sumitomo Special Metals Co., Ltd.,
2-15-17,Egawa, shimamototyou, Mishima-gun, Osaka, 618-0013,Japan,
I. Bolshakova, R. Holyaka, Magnetic Sensor Laboratory, Lviv Polytechnic National University,
1 Kotliarevsky Str., Lviv 79013, Ukraine

A Halbach cylinder is a magnetized cylinder composed of ferromagnetic material producing (in the idealized case) an intense magnetic field confined entirely within the cylinder with zero field outside. The cylinders can also be magnetized such that the magnetic field is entirely outside the cylinder, with zero field inside

Halbach arrays: used in synchrotrons and free electron lasers

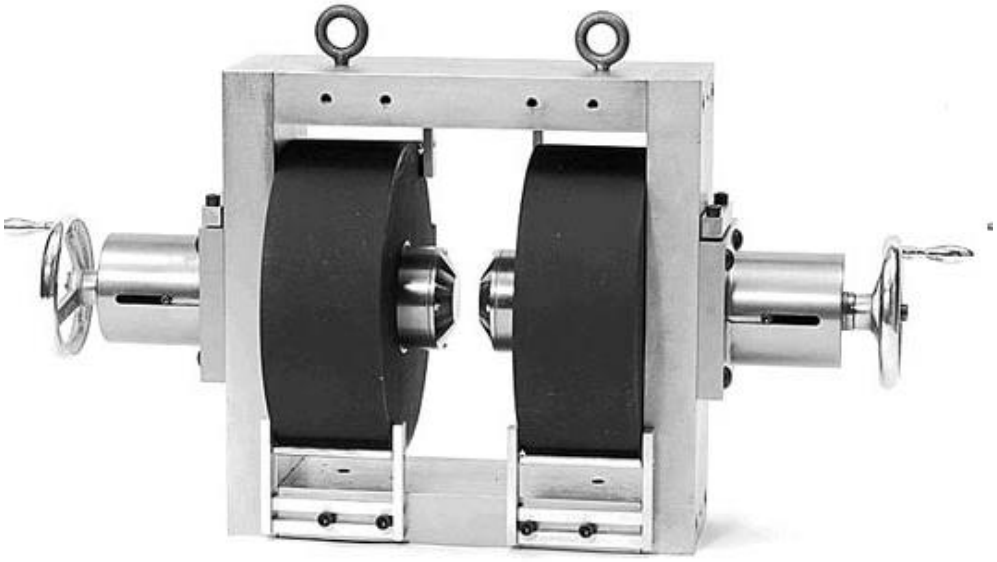
http://en.wikipedia.org/wiki/Halbach_array

Electromagnets

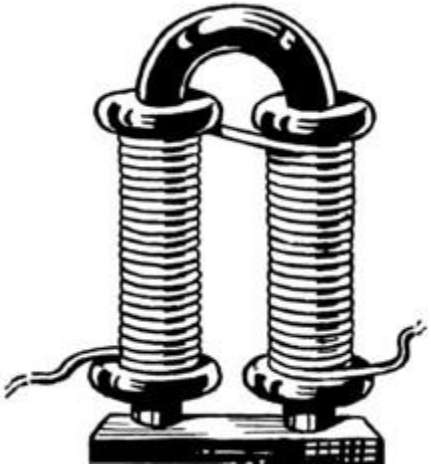


Biot-Savart law





Lab electromagnet



$$NI = B(\frac{L_{core}}{\mu} + \frac{L_{gap}}{\mu_0})$$

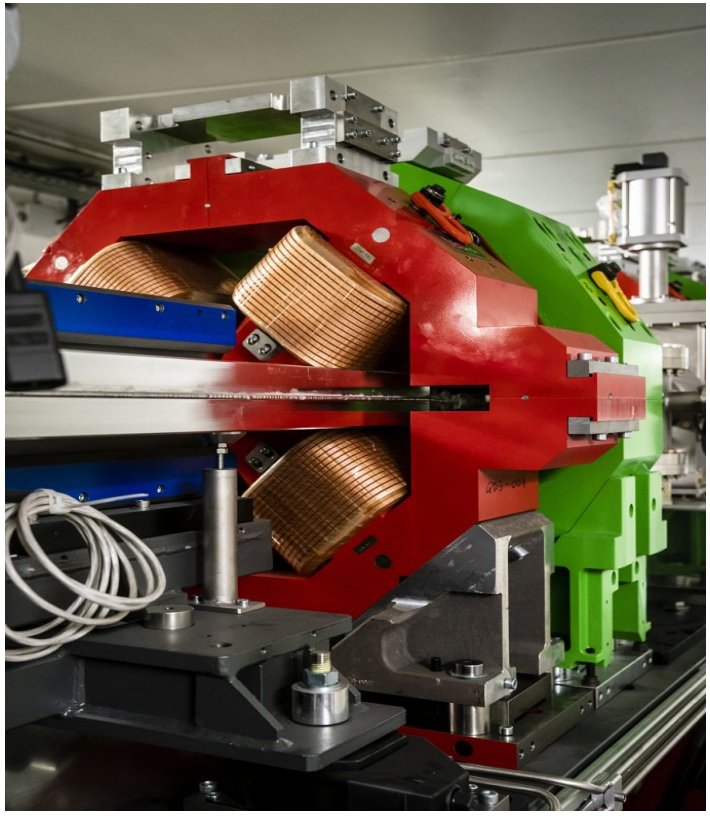
Electromagnets

Simple and straightforward

Field at room temperature

Reasonable fields (<3 Tesla)

No cryogens



Heavy and large

Small field volume

High field gradients

High power consumption

Need (water) cooling

Stability is determined by power supply

Electromagnets

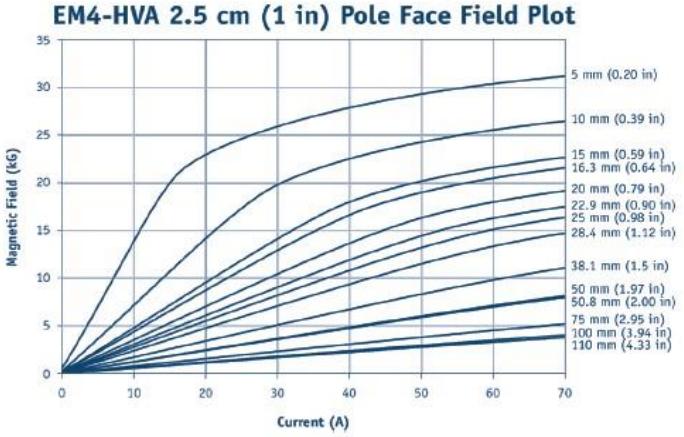


	EM4-HVA	EM7-HV
Air gap	Up to 11 cm (4.3 in) continuously variable	Up to 17.8 cm (7 in) continuously variable
Nominal coil resistance	0.25 Ω per coil (0.5 Ω total wired in series)	1.0 Ω per coil (0.5 Ω total wired in parallel)
Standard pole cap diameter	5.1 cm (2 in)	7.6 cm (3 in)
Optional pole cap diameter	2.5 cm, 7.6 cm, 10.2 cm (1 in, 3 in, 4 in)	15.2 cm (6 in)
Cooling water	Tap water or closed cooling system	
Water flow rate	7.6 L/min (2 gal/min)	11.4 L/min (3 gal/min)
Pressure drop	200 kPa (30 psi)	220 kPa (32 psi)
Water chiller cooling capacity	2.5 kW (8,530 BTU/h)	5.0 kW (17,060 BTU/h)
Water inlet temperature	15 °C to 25 °C (59 °F to 77 °F)	15 °C to 32 °C (59 °F to 89 °F)
Coil over temperature limit	70 °C (158 °F)	
Coil spacing, nominal	12 cm (4.75 in)	17.8 cm (7 in)
Coil size-width, nominal	12 cm (4.75 in)	13.2 cm (5.2 in)
Coil size-diameter, nominal	31.1 cm (12.25 in)	44.5 cm (17.5 in)
Current (maximum continuous operating)	±70 A per coil	±50 A per coil
Voltage	±35 V nominal	±50 V nominal
Maximum continuous input power	2.5 kVA nominal	5 kVA nominal
Lake Shore power supply (suggested)	Model 642	Model 665
	Size	
Height	38.7 cm at 40° angle (15.25 in) without stand	94 cm (37 in)
Width	84.5 cm (33.25 in)	101.6 cm (40 in)
Depth	50.8 cm (20 in)	66 cm (26 in)
Weight	201.9 kg (445 lb)	614 kg (1350 lb)
Shipping weight	215.5 kg (475 lb)	660 kg (1500 lb)
Shipping dimensions	95.5 cm × 58 cm × 56 cm (38 in × 23 in × 22 in)	86 cm × 122 cm × 119.4 cm (34 in × 48 in × 47 in)

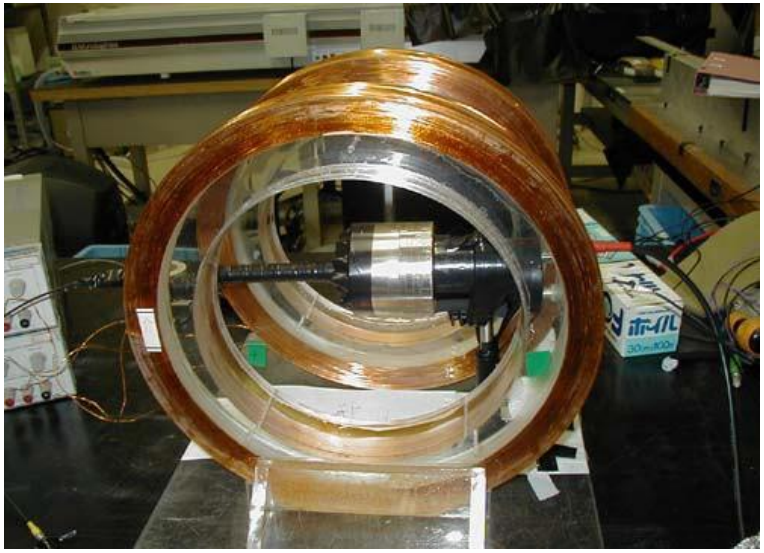
Typical Field Uniformity (EM4 Magnets)

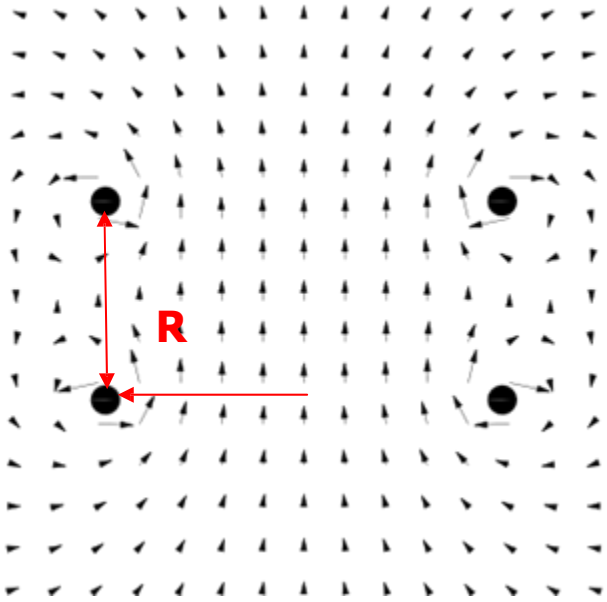
Magnet configuration		Uniformity	1% cylindrical volume	
Pole cap cm (in)	Air gap cm (in)	over 1 cm³	Diameter cm (in)	Length cm (in)
10 (4.0)	5.0 (2.0)	±0.15%	1.8 (0.7)	5.0 (2.0)
10 (4.0)	2.5 (1.0)	±0.05%	6.5 (2.5)	2.5 (1.0)
7.6 (3.0)	2.5 (1.0)	±0.06%	4.6 (1.8)	2.5 (1.0)
5.0 (2.0)	2.5 (1.0)	±0.35%	1.8 (0.7)	2.5 (1.0)
5.0 (2.0)	1.3 (0.5)	±0.16%	3.6 (1.4)	1.2 (0.5)

The third column gives uniformity over one cubic centimeter volume centered in the magnet gap. The last two columns give the cylindrical volume within which the magnetic field deviates by less than 1% from the central field. The cylindrical volume is coaxial with the magnet poles and centered in the gap.

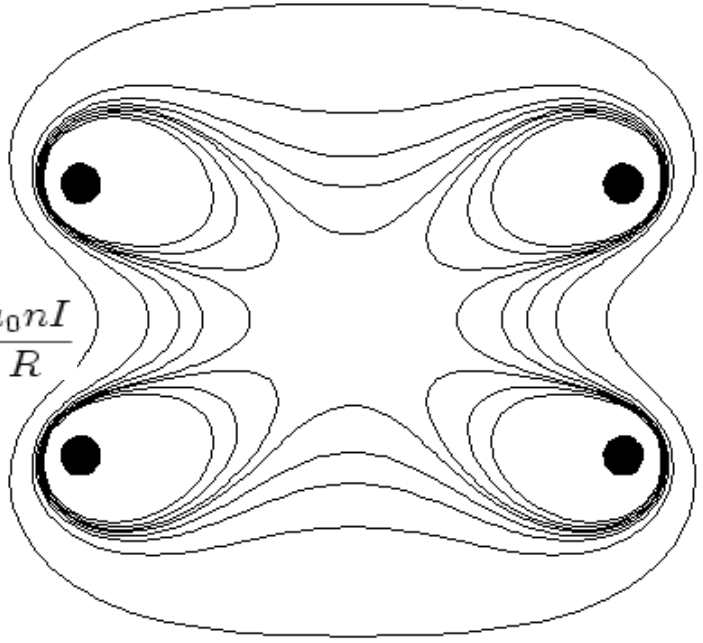


Helmholtz coil – uniform field in large volume

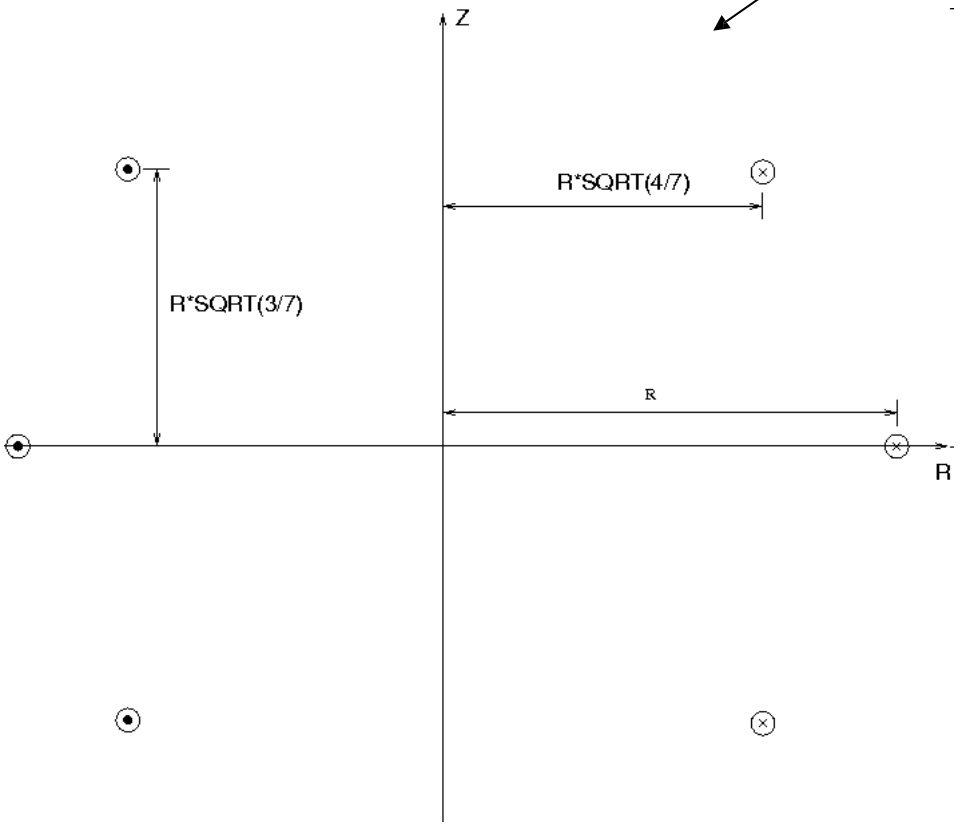


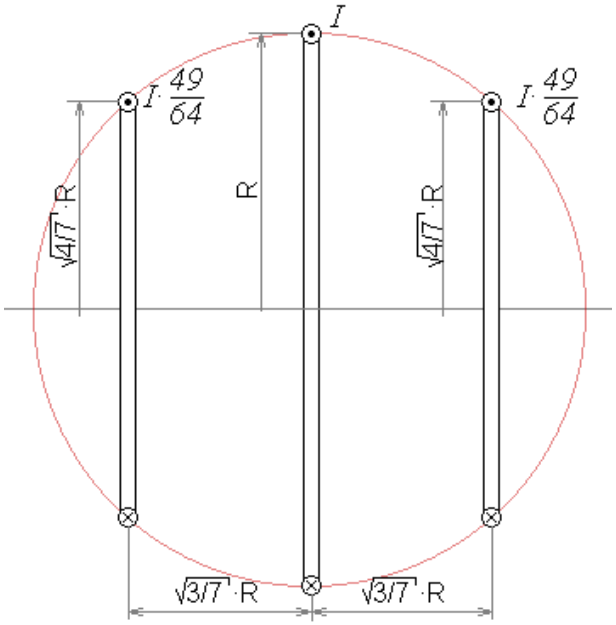


$$B = \left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 n I}{R}$$



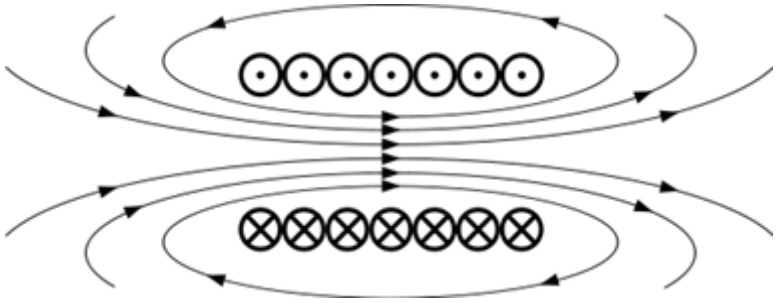
Maxwell coil – uniform field or uniform gradient

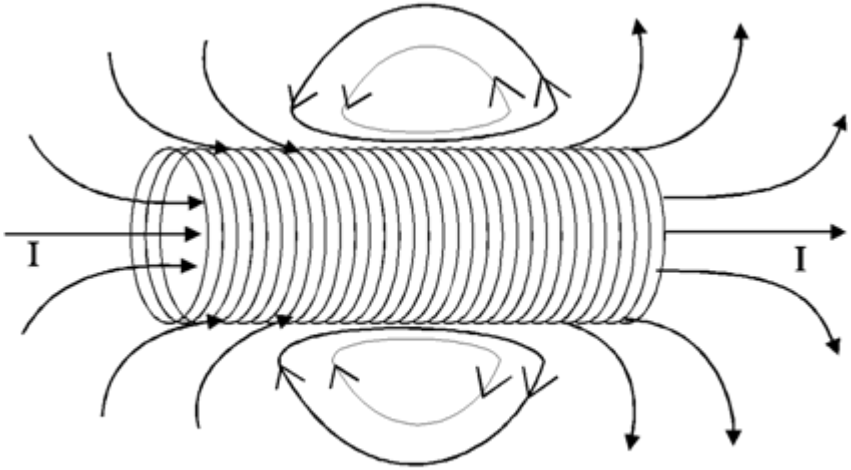




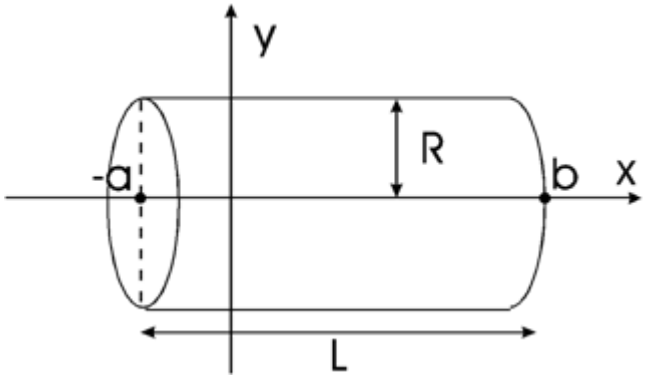
$nI_{small} = 49/69\ nI_{central}$

Solenoids

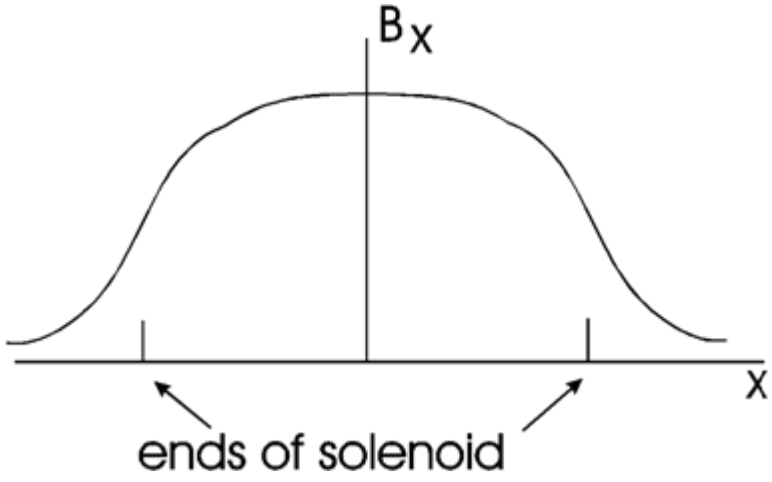




Field lines point in the direction South → North



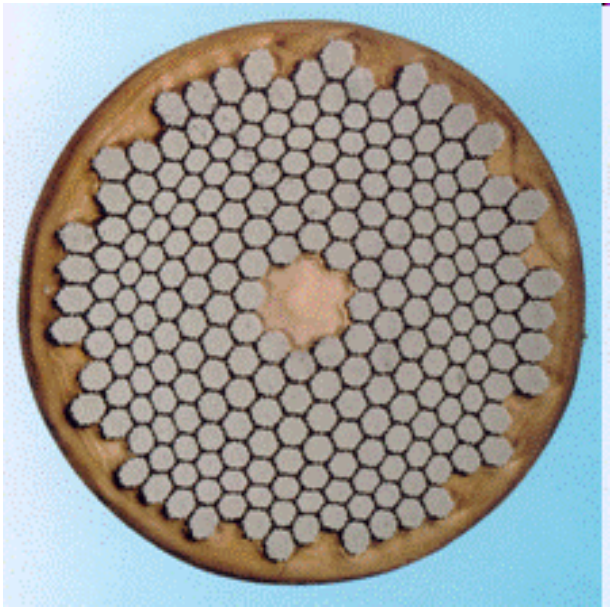
$$B = \frac{1}{2}\mu_0 nI\left(\frac{b}{\sqrt{b^2+R^2}} + \frac{a}{\sqrt{a^2+R^2}}\right)$$



Superconducting solenoids

Nb-Ti (T_c ~ 10 K, H_{c2} ~ 15 T)

Nb_3Sn (T_c ~ 18 K, H_{c2} ~ 30 T)



Multifilamentary cable

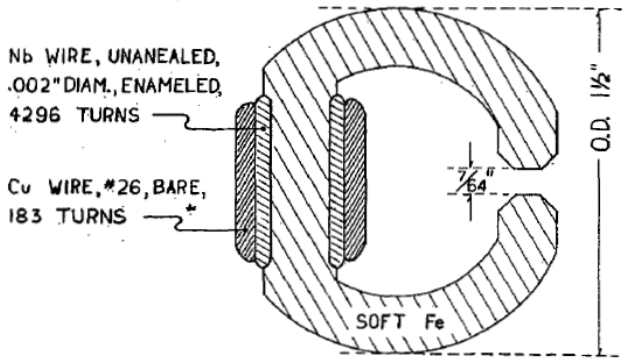
Copper (Cu-Ag, …) sheath

Serious metallurgical task

Epoxy-impregnation

First superconducting magnet

IEEE TRANSACTIONS ON MAGNETICS, VOL. MAG-23, NO. 2, MARCH 1987 390



Electromagnet with superconducting niobium windings, horizontal cross-section. Magnet constructed at University of Illinois in 1954.

0.71 T at 4.2 K

George Yntema

Phys. Rev. **98** (1955) 1197

occur at the same time. Increasing the magnetic field to several times the critical value produces no further change in the aforementioned properties of the metal.

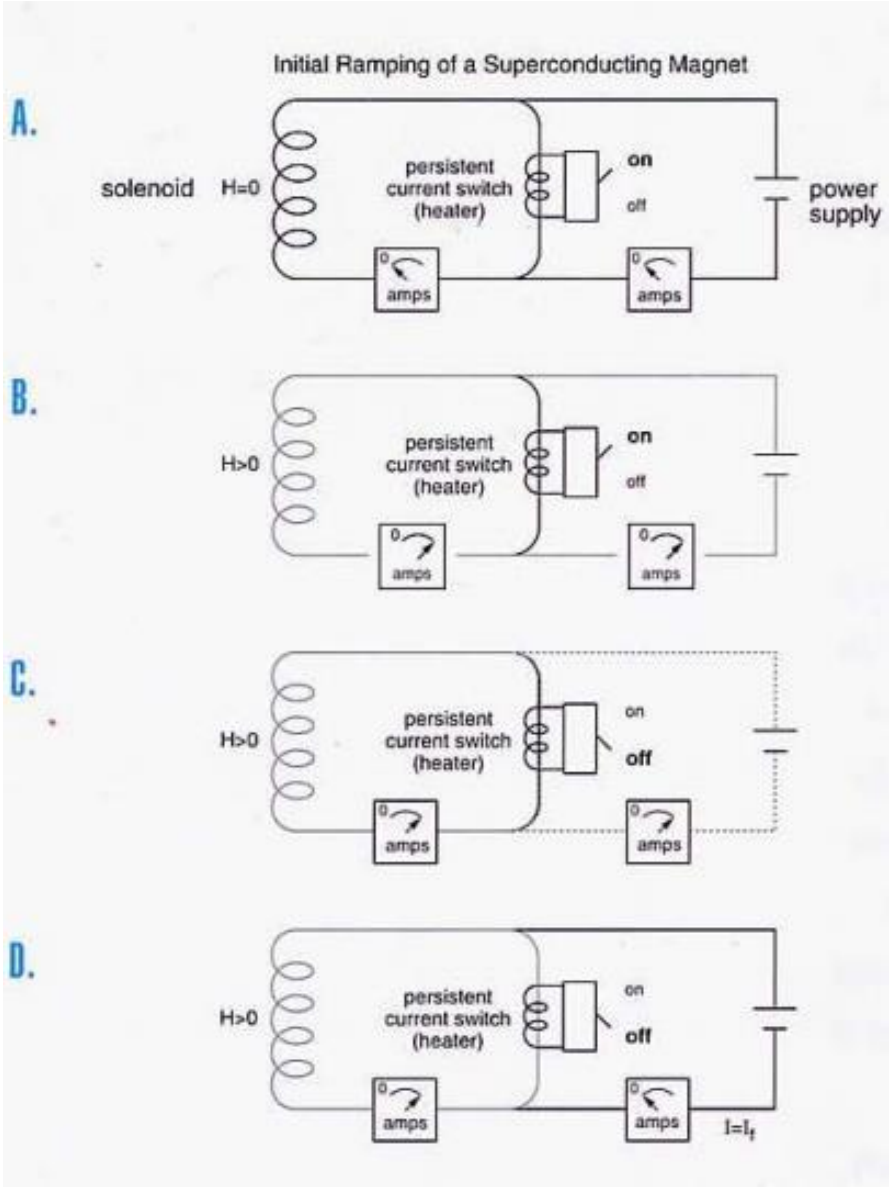
[1] H. E. Bommel, Phys. Rev. 96, 220 (1954).
[2] S. L. Quimby, Phys. Rev. 25, 558 (1925).

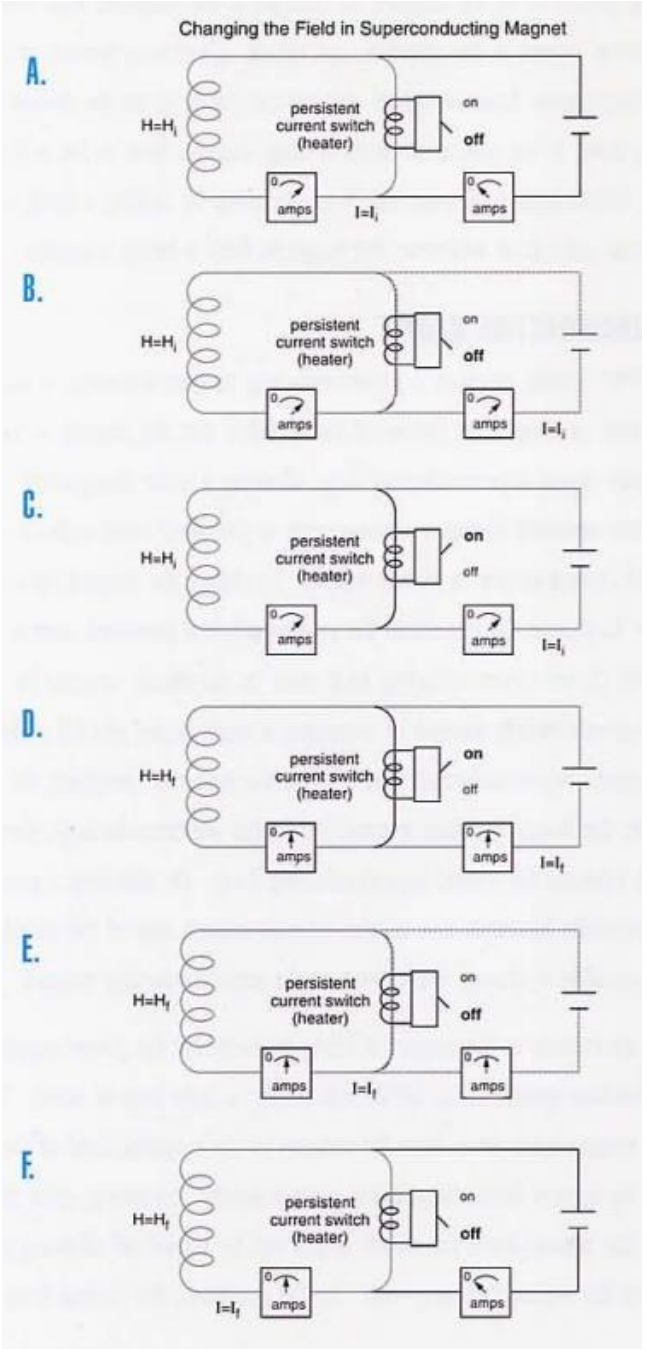
W8. Superconducting Winding for Electromagnet.* G. B. YNTEMA, *Cornell University.*—Superconducting niobium wire constitutes the winding of an electromagnet for use in a liquid helium bath. The model built has 4296 turns of 0.002 inch diam unannealed wire carrying 1.8 ampere and produces 7.1 kilogauss in a small gap. The wire remains superconducting while carrying a large current only so long as no segment of it is converted to the normal-conducting state; presumably such a resistive segment, when once started, grows because of the rapid production of heat within it. To avoid regions of concentrated heat production at the junctions of the Nb wire to copper leads, these junctions were spread out by spot welding the Nb at close intervals along several centimeters of its length to nickel foil, which was soldered to the copper. Particular samples of 0.0011 inch diam unannealed Nb at 4.2°K carried up to 9 ampere; 0.002 inch diam, 10 ampere; and 0.005 inch diam, 29 ampere. In an externally applied transverse magnetic field of 5 kilogauss at 1.7°K, 0.002 inch wire carried 1.5 ampere. Solenoids and iron-core magnets with such windings should be useful in various cryogenic experiments.

* Research done at the University of Illinois with support from the U. S. Atomic Energy Commission.

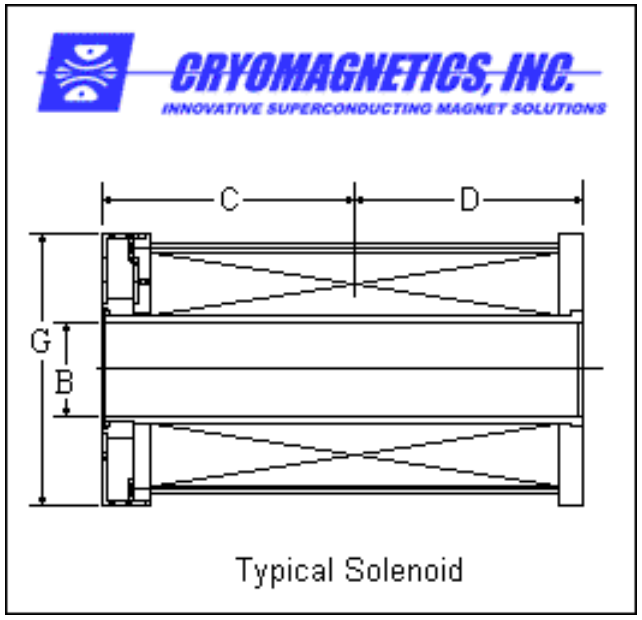
W9. On the Rotational Transition in Solid Hydrogen. K. TOMITA,* *Harvard University.*—Cooperative appearance of the rotation of hydrogen molecules, which is indicated both by proton resonance[1] and by specific heat,[2] is described by a semi-classical theory. The order of magnitude of the restriction to the rotation which is experienced by a molecule is estimated to be 4°K. The results derived from several different types of experiments, *viz.*, line shape of proton resonance, spin-lattice relaxation-time, and transition temperature of specific heat, are in reasonable agreement with each other. For the case of pure ortho-hydrogen, a calculation of the potential energy and the anisotropy based on intermolecular forces is carried out.

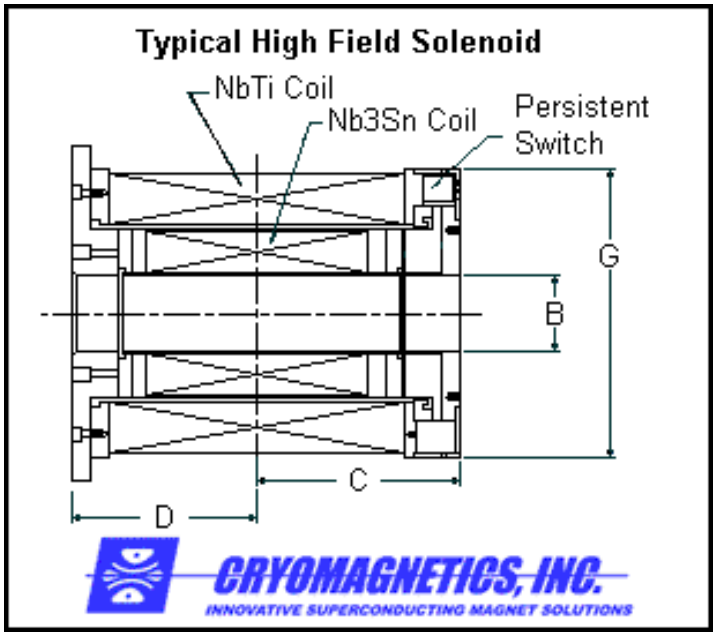
* On leave from the University of Kyoto.
[1] J. Hatton and B. V. Rollin, Proc. Roy. Soc. (London) A199, 222 (1949); R. Reif and E. M. Purcell, Phys. Rev. 91, 631 (1953).
[2] R. W. Hill and B. W. A. Ricketson, Phil. Mag. 45, 377 (1954).


Initial Ramping of a Superconducting Magnet
A.
solenoid
H=0
persistent
current switch
(heater)
on
off
power
supply
amps
amps
B.
H>0
persistent
current switch
(heater)
on
off
amps
amps
C.
H>0
persistent
current switch
(heater)
on
off
amps
amps
D.
H>0
persistent
current switch
(heater)
on
off
amps
amps
I=I_f


Changing the Field in Superconducting Magnet
A.
H=H_i
persistent
current switch
(heater)
on
off
amps
I=I_i
amps
B.
H=H_i
persistent
current switch
(heater)
on
off
amps
amps
I=I_i
C.
H=H_i
persistent
current switch
(heater)
on
off
amps
amps
I=I_i
D.
H=H_f
persistent
current switch
(heater)
on
off
amps
amps
I=I_f
E.
H=H_f
persistent
current switch
(heater)
on
off
amps
I=I_f
amps
F.
H=H_f
persistent
current switch
(heater)
on
off
I=I_f
amps
amps

Superconducting solenoids





Up to 9 T

Up to 20 T

Current up to ~100 A (current leads are important!)

Quench protection circuit (sometimes proprietary, in old days could use external shunt)

Superconducting solenoids

CRYOGENIC
CRYOGENIC LIMITED

Up to 22 T



High Field Magnet to 22 Tesla

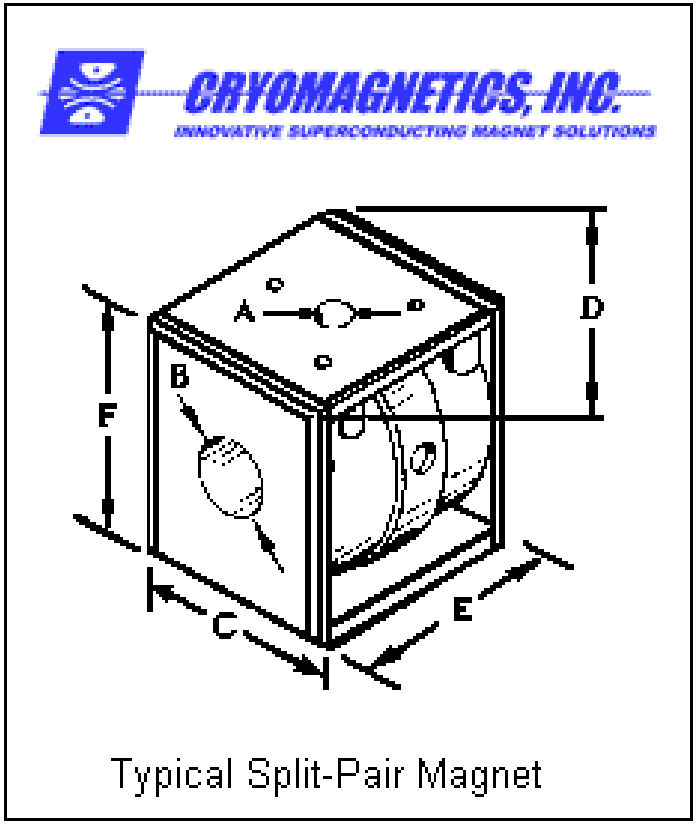
Specifications

Sample Specifications

Model	22T-30-H3	21T-30-H3
Central fields guaranteed at 4.2K	20.0 Tesla	19.0 Tesla
guaranteed at 2K	22.0 Tesla	21.0 Tesla
design at 4.2K	21.0 Tesla	20.0 Tesla
design at 2K	>23.0 Tesla	>22.0 Tesla
Maximum operating current	200A	200A
Inductance	100H	85H
Typical ramp rate with provided power supply		
0 to 18T	20 mins	18mins
18T to Bmax	l7mins	12mins
Central field homogeneity	----0 1% in 10mm----	
Clear central bore in 22T insert coil	30mm	
Clear bore in 20T main coil only	75mm	65mm
Overall diameter	350mm	335mm
Overall height	650mm	650mm

Nb_3Sn + HTS insert

Split pair magnets



Standard - up to 9T with 2" split access

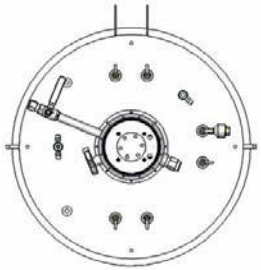
Angular dependencies, light/x-ray/neutron access

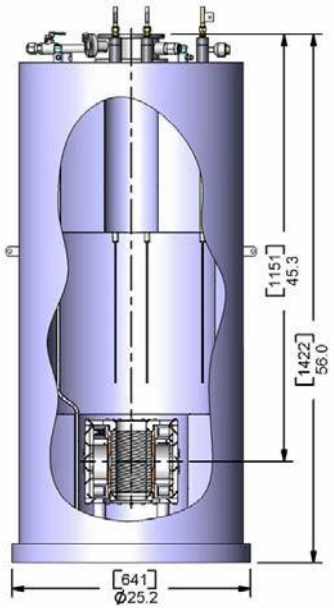
(less homogeneous field, more complex magnet design)

Vector magnet

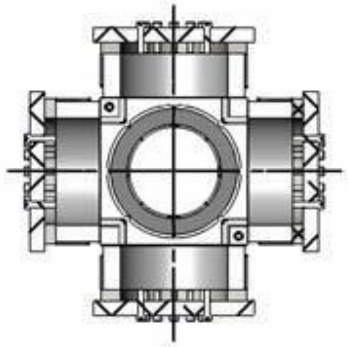
- expensive
- complex
- "small" fields

- no moving parts
- precise value/direction of the field



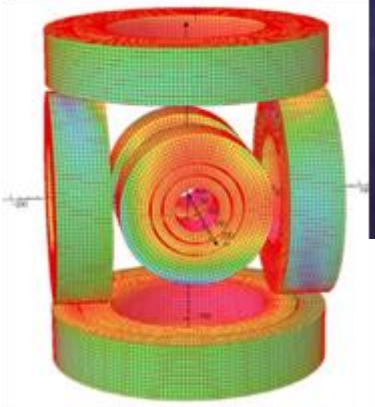


1 Tesla



1 Tesla

7 Tesla
vertical field





Superconducting magnets – lambda plate

Magnet at 2.2 K, >10% increase in max. field

(think of lambda-plate as of VTI with a valve that is cooling the magnet)

Can also pump on He-bath but this is too He-consuming.

Superconducting magnets

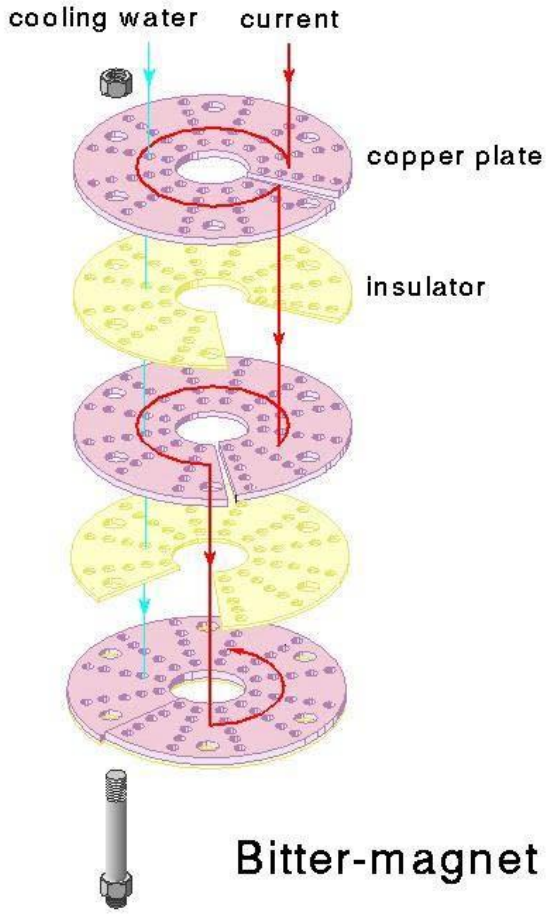
Affordable and compact high magnetic fields.
Workhorse in CMP laboratory.

Need cryogens

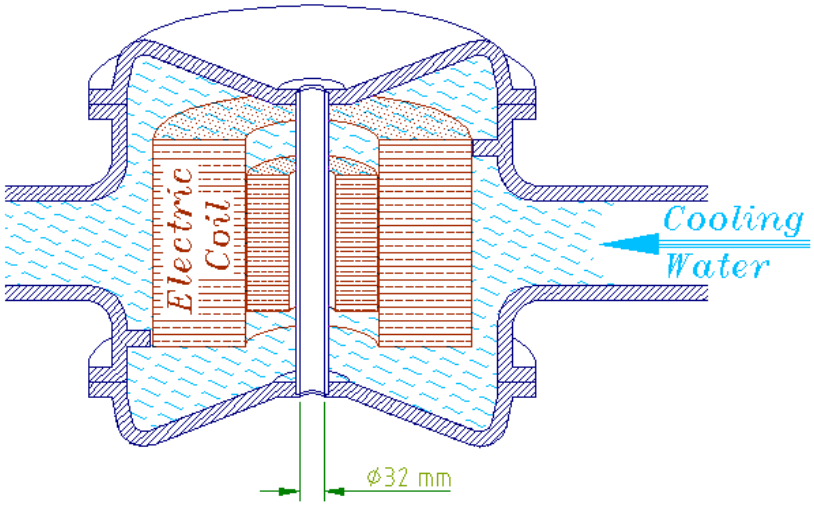
Flux jumps

Not so trivial if T > 300 K desired

Bitter resistive magnets



Bitter magnets are constructed of circular conducting metal plates and insulating spacers stacked in a helical configuration, rather than coils of wire. The current flows in a helical path through the plates. This design was invented in 1933 by American physicist, Francis Bitter.



NHMFL – Tallahassee – 35 T

NHMFL – 45T hybrid magnet

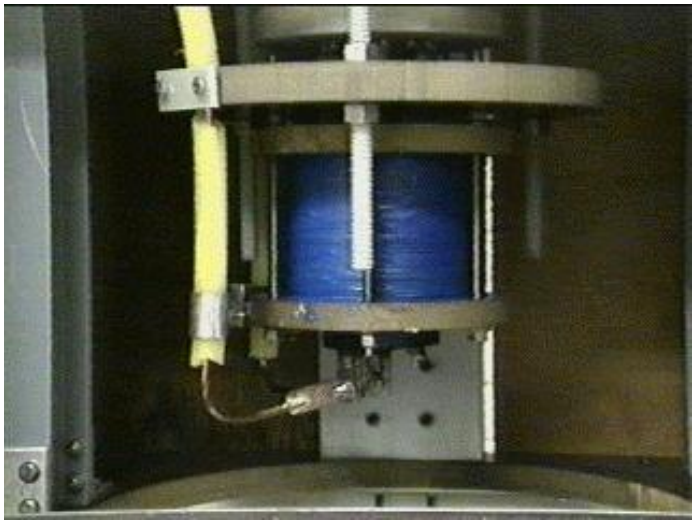


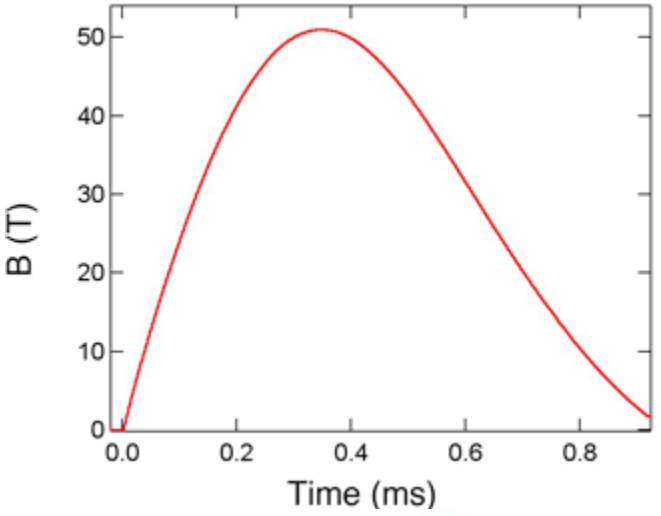
Strength	45 tesla
Type	Hybrid
Bore size	32 mm (~1.25 inches)
Online since	December 1999
Cost	$14.4 million
Weight	31,752 kg (35 tons)
Height	6.7 meters (22 feet)
Operating temperature	-271 ° C (-456 ° F)
Water used per minute	15,142 liters (4,000 gallons)
Power required	33 MW

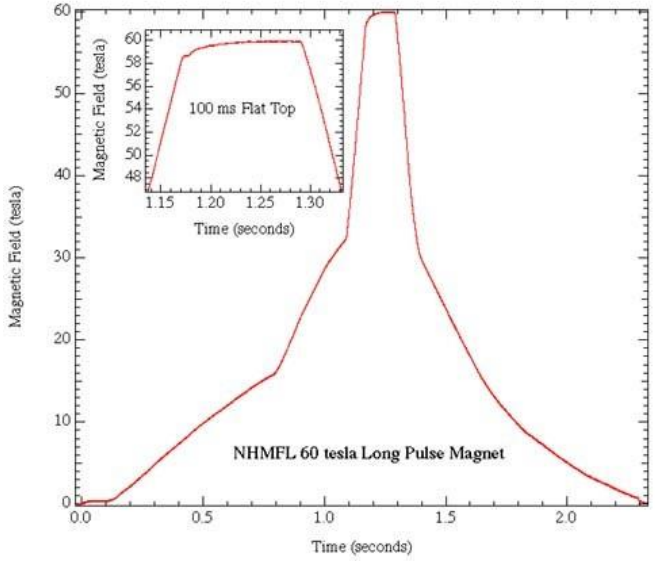
33.5 T resistive + 11.5 T superconducting

Operation cost (full field) ~ 4,000 $/h

Pulse field magnets







~ 1 shot every 45 min.

Fast data acquisition

Heating

Fast processes

Mechanical strength of the coil is an issue

Pulse field magnets – NHMFL – Los Alamos

Capacitor Bank-Driven Magnets		
Field	**Duration**	**Bore**
50 T Short Pulse	25 msec	24 mm
50 T Mid-Pulse	400 msec	15 mm
40 T Mid-Pulse	400 msec	24 mm
65 T Short Pulse	25 msec	15 mm
60 T Short Pulse	40 msec	9.8 mm
300 T Single Turn	6 µsec	10 mm

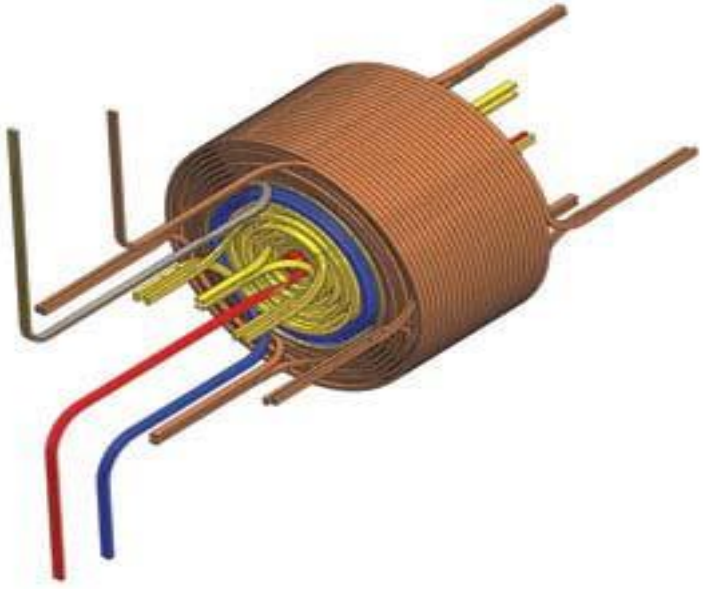
Generator-Driven and Multiplex Magnets		
Field	**Duration**	**Bore**
60 T Controlled Waveform	100 msec	15 mm
100 T Multi-Shot (operational to 90 T)	25 msec	15 mm

magneto-optics (IR through UV), magnetization and magneto-transport from 350 mK to 300K; GHz conductivity, MHz conductivity, pulse echo ultra-sound spectroscopy, magnetoconductivity and heat capacity.

100 T pulse magnet – NHMFL - LANL





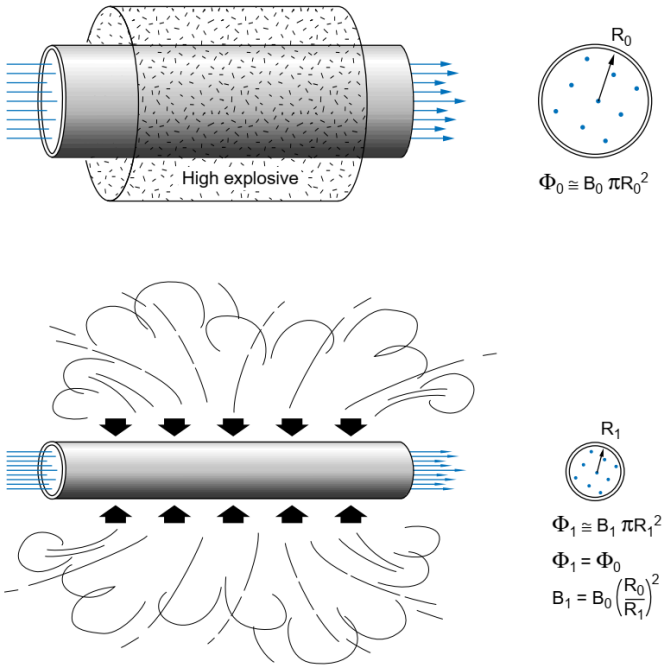


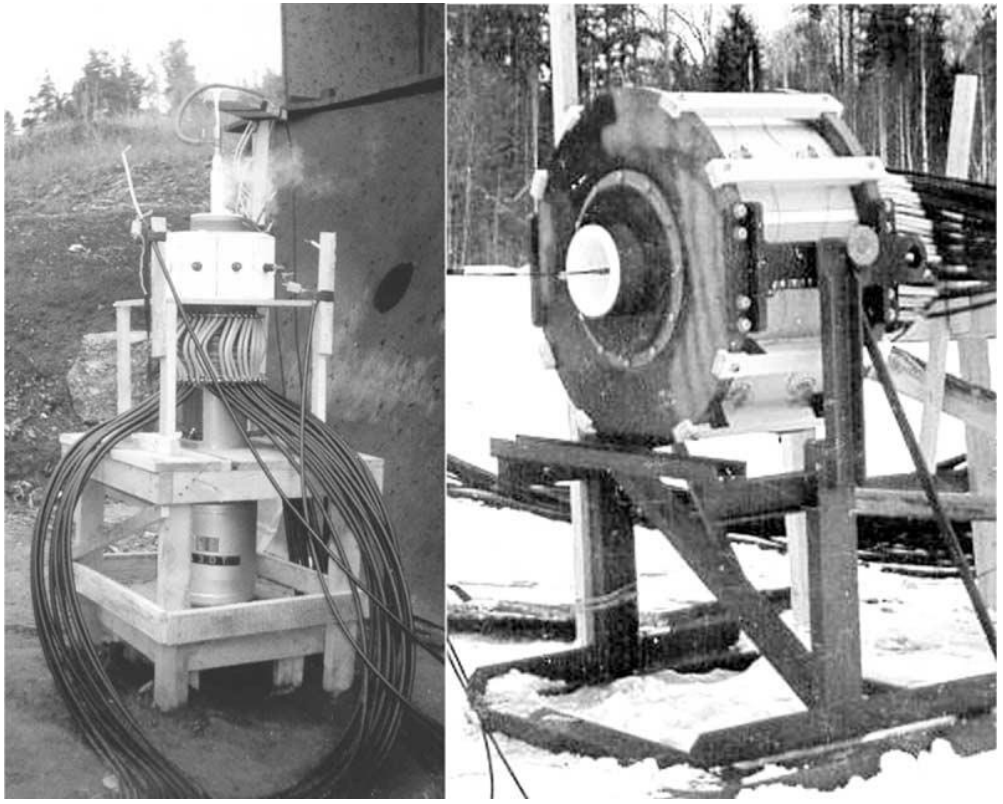
Destructive pulse field magnets

300 T single turn – Los Alamos

Electromagnetic flux compression (~750 T) - ISSP

Explosive flux compression - ~ 2800 T – Sarov (Russia)





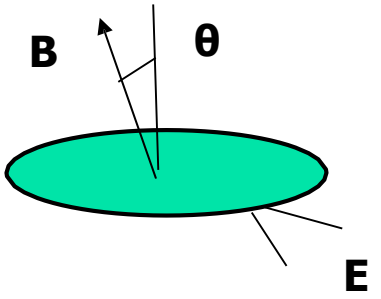
Magnetic field measurements

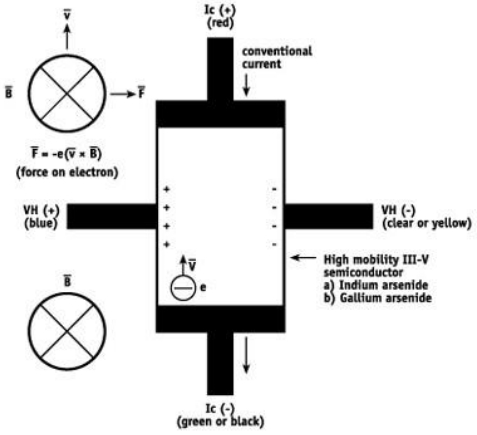
(*) Well defined geometry coil – can calculate.

(**) Faraday's law: $E = - dB/dT \cdot nS \cdot \cos\theta$ – several assumptions: constant area S, measurable θ, fast, accurate electronics. Good for pulsed fields.

(***) Hall probes – linear in field. 2DEG – very sensitive but at low temperatures/high fields QHE and/or SdH are observed. Other (e.g. III – V) semiconductors. Semimetals (Bi, …). Either purchase (LakeShore, GMW, …) or DIY – if you have even primitive thin film technology. **NB:** *Temperature dependence, angle with magnetic field (but – can serve as angle sensor), linearity.* Hall arrays, multi-axes sensors.

(****) Standards (susceptibility of pure Pd for MPMS)





Miniature Hall-probe Array

